$$\mathrm{var}\ \mathrm{vz}\,[\langle t\rangle] = t$$

$$\mathrm{var}\,(\mathrm{vs}\ x)\,[\langle t\rangle] = \mathrm{var}\ x$$

$$(\mathrm{v1} + \mathrm{v0})\,[\langle \mathrm{m2}\rangle]\,[\langle \mathrm{m1}\rangle^{+}] =$$

$$\mathrm{v1}\,[\langle \mathrm{m2}\rangle]\,[\langle \mathrm{m1}\rangle^{+}] + (\mathrm{m2}\,[\langle \mathrm{m1}\rangle^{+}])$$

$$= \mathrm{v0}\,[\langle \mathrm{m1}\rangle^{+}] + \mathrm{m2} =$$

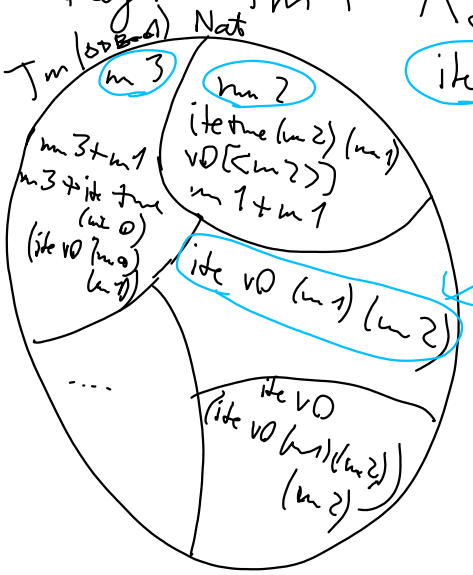
$$\mathrm{v0} + \mathrm{m2}$$

Razor: $\mathrm{Tm}\ A \underset{\lceil - \rceil}{\overset{\mathrm{norm}}{\cong}} \mathrm{Nf}\ A$ (compl, stab.)

$\mathrm{Nf}\ \mathrm{Bool} := \mathbb{Z}$

$\mathrm{Nf}\ \mathrm{Nat} := \mathbb{N}$

Def. $\mathrm{Tm}\ \Gamma\ A \underset{\lceil - \rceil}{\overset{\mathrm{norm}}{\cong}} \mathrm{Nf}\ \Gamma\ A$ (compl, stab)



$\mathrm{ite}\ \mathrm{v0}\ (\mathrm{m1} + \mathrm{m2})\ (\mathrm{m1})$ — norm neutrális

neutrális

$\mathrm{istero}\ \mathrm{v0}$

$\mathrm{istero}\ (\mathrm{ite}\ \mathrm{v0}\ (\mathrm{m0})\ (\mathrm{m0}))$

$\mathrm{v0} + \mathrm{v1}$

$\mathrm{v0} + (\mathrm{m1} + \mathrm{m2})$

neutrális

$\mathrm{Nf} : \mathrm{Con} \to \mathrm{Ty} \to \mathrm{Set}$

$\mathrm{nm} : \mathbb{N} \to \mathrm{Nf}\ \Gamma\ \mathrm{Nat}$

$\mathrm{true}, \mathrm{false} : \mathrm{Nf}\ \Gamma\ \mathrm{Bool}$

$\mathrm{neu} : \mathrm{Ne}\ \Gamma\ A \to \mathrm{Nf}\ \Gamma\ A$

$\mathrm{Ne} : \mathrm{Con} \to \mathrm{Ty} \to \mathrm{Set}$

$\mathrm{var} : \mathrm{Var}\ \Gamma\ A \to \mathrm{Ne}\ \Gamma\ A$

$\mathrm{ite} : \mathrm{Ne}\ \Gamma\ \mathrm{Bool} \to \mathrm{Nf}\ \Gamma\ A \to \mathrm{Nf}\ \Gamma\ A \to \mathrm{Ne}\ \Gamma\ A$

$\mathrm{istero} : \mathrm{Ne}\ \Gamma\ \mathrm{Nat} \to \mathrm{Ne}\ \Gamma\ \mathrm{Bool}$

$_+_ : \mathrm{Ne}\ \Gamma\ \mathrm{Nat} \to \mathrm{Ne}\ \Gamma\ \mathrm{Nat} \to \mathrm{Ne}\ \Gamma\ \mathrm{Nat}$

$_+_ : \mathrm{Ne}\ \Gamma\ \mathrm{Nat} \to \mathbb{N} \to \mathrm{Ne}\ \Gamma\ \mathrm{Nat}$

$_+_ : \mathbb{N} \to \mathrm{Ne}\ \Gamma\ \mathrm{Nat} \to \mathrm{Ne}\ \Gamma\ \mathrm{Nat}$

"vO + vO": Nf (∅▷Nat) Nat

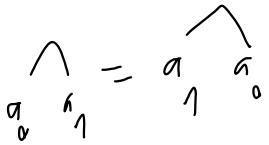
neu (var vO + var vO)

neu (var vO + neu (var vO))

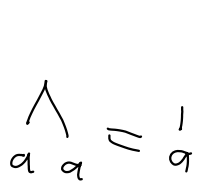
lista $[a_0, a_1, a_2, \dots]$

$[a_0, a_0, a_0, \dots a_1, a_1, \dots, a_2, a_2 \dots]$

zsák (bag) multiset



halmaz



Raptor (mincs Gn, Sub, Var)

Kombinátor logika (SK kalkulus)

Ty : Set

Tm : Ty → Set

_⇒_ : Ty → Ty → Ty

_$_ : Tm (A⇒B) → Tm A → Tm B

K : Tm (A ⇒ B ⇒ A)

Kβ : K $ u $ v = u

S : Tm ((A⇒B⇒C) ⇒ (A⇒B) ⇒ A ⇒ C)

Sβ : S $ t $ u $ v = (t $ v) $ (u $ v)

t $ u $ v = (t $ u) $ v

A ⇒ B ⇒ C = A ⇒ (B ⇒ C)

‖

(A ⇒ B) ⇒ C

2-param. fgv.

1-param. fgv.

I : Tm (A ⇒ A)

I := S $ K $ K

I $ u ≟ u

‖ def

S $ K $ K $ u

‖ Sβ

K $ u $ (K $ u)

‖ Kβ

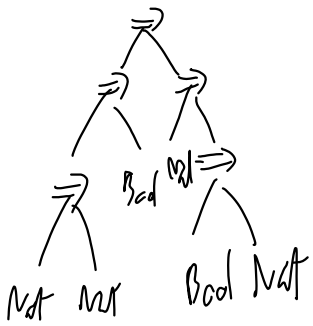
u

STT simple type theory, egyszerű típusos λ-kalkulus Bool, Nat

Def + _⇒_ : Ty → Ty → Ty

des _$_ : Tm Γ (A ⇒ B) → Tm Γ A → Tm Γ B

$[] : (t $ u)[σ] = (t[σ]) $ (u[σ])



kon lam : Tm (Γ▷A) B → Tm Γ (A ⇒ B)

kötő op.

lam[] : (lam t)[σ] = lam (t[σ+])

σ : Sub Δ Γ

Sub (Δ▷A) (Γ▷A)

⇒β : lam t $ u = t[⟨u⟩]

⇒η : (t : Tm Γ (A ⇒ B)) → t = lam (t[p] $ v0)

$$\frac{\dfrac{\dfrac{t : \mathrm{Tm}\ \Gamma\ (A \Rightarrow B)}{t[p] : \mathrm{Tm}\ (\Gamma \triangleright A)\ (A \Rightarrow B)} \quad \dfrac{v0 : \mathrm{Tm}\ (\Gamma \triangleright A)\ A}{}}{t[p] \$ v0 : \mathrm{Tm}\ (\Gamma \triangleright A)\ B}}{\mathrm{lam}\ (t[p] \$ v0) : \mathrm{Tm}\ \Gamma\ (A \Rightarrow B)}$$

p : Sub (Γ▷A) Γ

t = λx. t x

β : számítási szab (kom)

η : egyértelmiség kom (def)

$\mathrm{add2} : \mathrm{Tm}\ \diamond\ (\mathrm{Nat} \Rightarrow \mathrm{Nat})$

$\mathrm{add2} = \mathrm{lam}\ (\mathrm{v0} + \mathrm{tm}\ 2)$ $\quad (\lambda x.\, x+2)$

$: \mathrm{Tm}\ (\diamond \triangleright \mathrm{Nat})\ \mathrm{Nat}$

$\Rightarrow \beta : \mathrm{lam}\ t \$ u = t[\langle u \rangle]$

$+[]$

$$\mathrm{add2}\ \$\ \mathrm{tm}\ 4 = \mathrm{lam}(\mathrm{v0} + \mathrm{tm}\ 2)\ \$\ \mathrm{tm}\ 4 \overset{\beta}{=} (\mathrm{v0} + \mathrm{tm}\ 2)[\langle \mathrm{tm}\ 4 \rangle] =$$

$$(\mathrm{v0}[\langle \mathrm{tm}\ 4 \rangle]) + (\mathrm{tm}\ 2\ [\langle \mathrm{tm}\ 4 \rangle]) \overset{\mathrm{tm}[]}{=} \mathrm{v0}[\langle \mathrm{tm}\ 4 \rangle] + \mathrm{tm}\ 2 \overset{\mathrm{vz}[\langle\rangle]}{=} \mathrm{tm}\ 4 + \mathrm{tm}\ 2 \overset{+\beta}{=} \mathrm{tm}\ 6$$

Def: $\mathrm{Fun}\ \Gamma\ A\ B := \mathrm{Tm}\ (\Gamma \triangleright A)\ B$

$_\$_ : \mathrm{Tm}\ \Gamma\ A\ B \to \mathrm{Tm}\ \Gamma\ A \to \mathrm{Tm}\ \Gamma\ B$

$\mathrm{hom} : \mathrm{Tm}\ \diamond\ ((\mathrm{Nat} \Rightarrow \mathrm{Nat}) \Rightarrow \mathrm{Nat} \Rightarrow \mathrm{Nat})$

$\lambda f\ x.\, f(f(f\,x))$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dots}{\mathrm{v1} : \mathrm{Tm}\ \Gamma\ (\mathrm{N} \Rightarrow \mathrm{N})} \quad \dfrac{\dfrac{\dots}{\mathrm{v1} : \mathrm{Tm}\ \Gamma\ (\mathrm{N} \Rightarrow \mathrm{N})} \quad \dfrac{\dfrac{\dots}{\mathrm{v1} : \mathrm{Tm}\ \Gamma\ (\mathrm{N} \Rightarrow \mathrm{N})} \quad \dfrac{}{\mathrm{v0} : \mathrm{Tm}\ \Gamma\ \mathrm{N}}}{\mathrm{v1}\ \$\ \mathrm{v0} : \mathrm{Tm}\ \Gamma\ \mathrm{N}}}{\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ \mathrm{v0}) : \mathrm{Tm}\ \Gamma\ \mathrm{N}}}{\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ \mathrm{v0})) : \mathrm{Tm}\ (\diamond \triangleright \mathrm{N} \Rightarrow \mathrm{N} \triangleright \mathrm{N})\ \mathrm{N}}}{\mathrm{lam}(\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ \mathrm{v0}))) : \mathrm{Tm}\ (\diamond \triangleright \mathrm{N} \Rightarrow \mathrm{N})\ (\mathrm{N} \Rightarrow \mathrm{N})}}{\mathrm{lam}(\mathrm{lam}(\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ (\mathrm{v1}\ \$\ \mathrm{v0})))) : \mathrm{Tm}\ \diamond\ ((\mathrm{N} \Rightarrow \mathrm{N}) \Rightarrow \mathrm{N} \Rightarrow \mathrm{N})}$$

$=: \Gamma$

Curry-Zé